DBMS - UNIT 5: NORMALIZATION

Prof. A. Mehta | Database Management Systems

Why normalise? To remove redundancy and the insertion, update and deletion anomalies that come with it.

Functional dependency (FD): X -> Y means the value of X determines the value of Y.
Example: RollNo -> Name. Closure of attributes X+ gives everything X determines; used to find candidate keys.
Armstrong's axioms: reflexivity, augmentation, transitivity.

First normal form (1NF): every attribute holds atomic values, no repeating groups.
Second normal form (2NF): 1NF + no partial dependency of a non-prime attribute on part of a candidate key.
Third normal form (3NF): 2NF + no transitive dependency of a non-prime attribute on a key.
For every FD X -> A: X is a superkey, or A is a prime attribute.

UNIT 5: NORMALIZATION (contd.)

Boyce-Codd normal form (BCNF): for every non-trivial FD X -> Y, X must be a superkey.
BCNF is stricter than 3NF. Every BCNF relation is in 3NF but not the other way round.

Decomposition properties
- Lossless join: joining the decomposed tables gives back exactly the original table. Test: the common attributes must be a key of at least one of the two tables.
- Dependency preservation: every FD can be checked inside a single table.
3NF decomposition always achieves both. BCNF decomposition is always lossless but may not preserve dependencies.

Worked example: R(A, B, C) with A -> B, B -> C. Key = A. B -> C is transitive, so R is not in 3NF.
Decompose into R1(A, B) and R2(B, C): both in BCNF, lossless join, dependencies preserved.

Denormalisation: deliberately adding redundancy back for faster reads in reporting systems.